



Department of Electrical & Electronic Engineering
Rajshahi University of Engineering & Technology

LAB Report

Course Code : **EEE 3100**

Course Title : **Electronics Shop Practice**

Experiment No. : 01

Experiment Name : Soldering & Testing of Astable Multivibrator

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Contents

1	Experiment No. 01	2
2	Experiment Title	2
3	Objectives	2
4	Theory	2
4.1	Astable Multivibrator Using 555 Timer	2
4.1.1	Working Principle	3
4.2	Time Period and Frequency Calculations	3
4.3	Characteristics of 555 Astable Multivibrator	4
5	Necessary Equipments	4
6	Circuit Diagram	4
7	Experimental Setup	5
8	Theoretical Result	5
9	Discussion	6
10	Conclusion	6

1 Experiment No. 01

2 Experiment Title

Soldering & Testing of an Astable Multivibrator

3 Objectives

1. To study the operation and observe the wave forms of Astable Multivibrator
2. To Design an Astable Multivibrator on PCB board and to generate a square wave

4 Theory

A Printed Circuit Board (PCB) is a rigid structure that houses electrical circuitry made up of embedded metal traces, which connect various electronic components to form a complete circuit. One fundamental circuit configuration used in electronic applications is the astable multivibrator. An astable multivibrator is an oscillator circuit that continuously switches between two quasi-stable states without requiring an external triggering signal. Due to its ability to generate a continuous square wave output, it is also referred to as a free-running multivibrator.

4.1 Astable Multivibrator Using 555 Timer

The 555 timer IC is a highly versatile component used for timing and waveform generation. When configured as an astable multivibrator, it functions as an oscillator, generating a continuous square wave output. This configuration allows the circuit to make automatic and successive transitions from one quasi-stable state to another without the need for an external input.

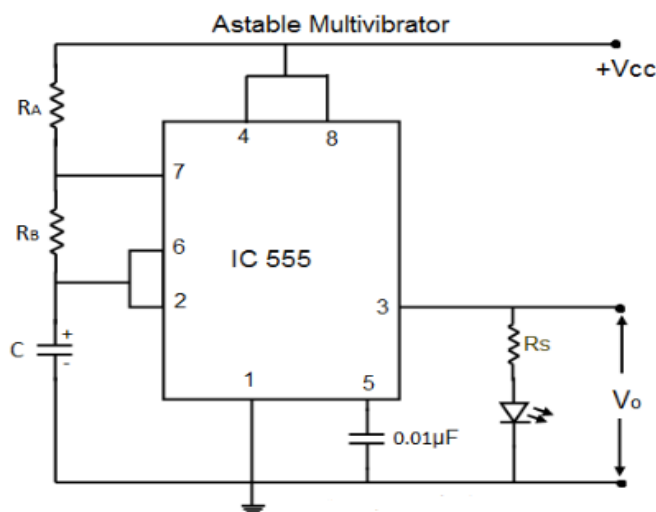


Figure 1.1: Astable Multivibrator Using 555 Timer IC

4.1.1 Working Principle

In a 555 timer astable circuit, pin 2 (Trigger) and pin 6 (Threshold) are connected together, allowing the circuit to retrigger itself on each cycle. This ensures continuous oscillation without external intervention.

The operation of the circuit can be explained in two phases (according to fig. 1.2):

- **Charging Phase:**

- The capacitor C begins charging through the resistors R_1 and R_2 .
- The voltage across the capacitor increases until it reaches $\frac{2}{3}V_{cc}$.
- The upper comparator in the 555 timer is activated, switching the internal flip-flop and turning the discharge transistor ON.

- **Discharging Phase:**

- Once the flip-flop is triggered, the capacitor C starts discharging through R_2 , as the other side of R_2 is connected to the discharge pin 7.
- The capacitor voltage decreases until it reaches $\frac{1}{3}V_{cc}$.
- The lower comparator is activated, resetting the flip-flop and turning the discharge transistor OFF, restarting the charging cycle.

This repeated charging and discharging cycle generates a square wave output at pin 3 (Output) of the 555 timer. The frequency and duty cycle of the output waveform are determined by the values of R_1 , R_2 , and C .

4.2 Time Period and Frequency Calculations

The total time period T of the square wave output is given by:

$$T = T_{HIGH} + T_{LOW} \quad (1)$$

Where:

- Time for HIGH state (T_{HIGH}):

$$T_{HIGH} = 0.693(R_1 + R_2)C \quad (2)$$

- Time for LOW state (T_{LOW}):

$$T_{LOW} = 0.693(R_2)C \quad (3)$$

The frequency (f) of oscillation is:

$$f = \frac{1}{T} = \frac{1.44}{(R_1 + 2R_2)C} \quad (4)$$

The duty cycle (D) is given by:

$$D = \frac{T_{HIGH}}{T} \times 100 = \frac{(R_1 + R_2)}{(R_1 + 2R_2)} \times 100 \quad (5)$$

4.3 Characteristics of 555 Astable Multivibrator

- Generates a continuous square wave output.
- The frequency and duty cycle depend on R_1 , R_2 , and C .
- Independent of supply voltage variations.
- Commonly used in pulse generation, LED blinking circuits, clock pulse generation, and waveform generation.

5 Necessary Equipments

1. PCB
2. Soldering iron
3. Resistor ($100K\Omega \times 1$, $10K\Omega \times 1$)
4. Capacitor ($10\mu F \times 2$)
5. LED
6. Potentiometer
7. DC supply
8. Jumper wires

6 Circuit Diagram

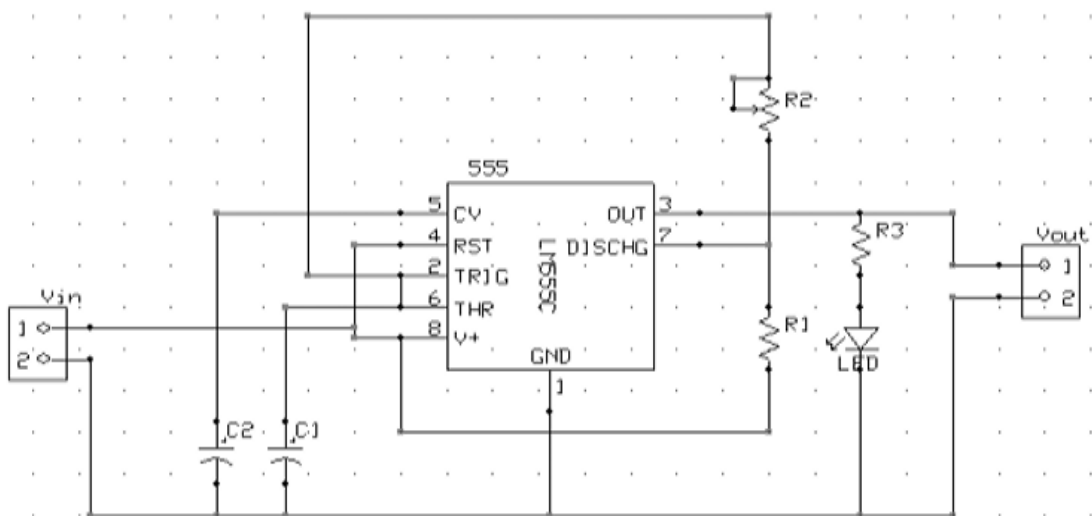


Figure 1.2: Circuit Diagram of Astable Multivibrator

7 Experimental Setup

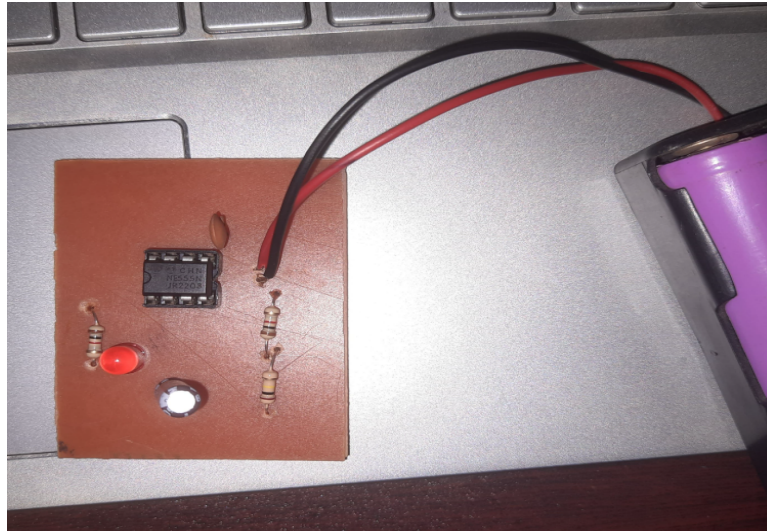


Figure 1.3: PCB Setup (Front Side)

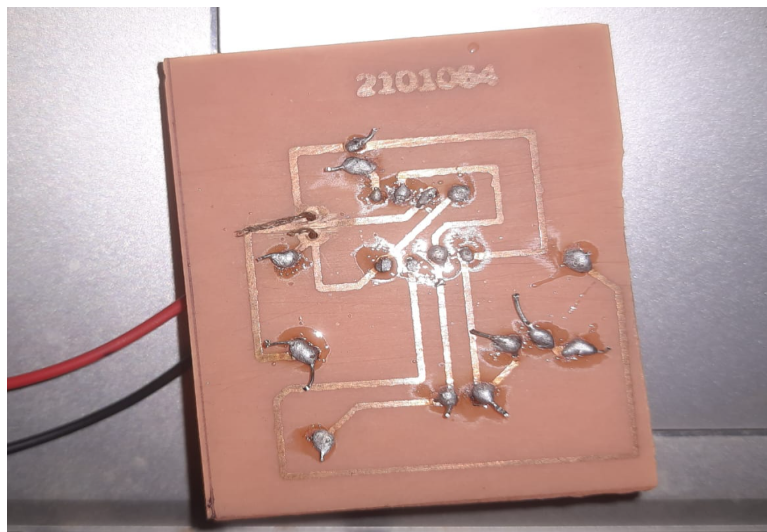


Figure 1.4: PCB Setup (Soldered Side)

8 Theoretical Result

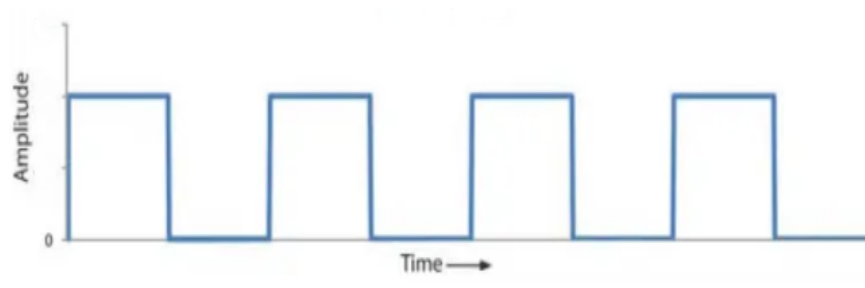


Figure 1.5: Output Waveshape of Astable Multivibrator (Theoretical)

9 Discussion

In this lab, implementing an astable multivibrator by using PCB was the main objective. As per the experiment, the output was to generate a square wave with a specific frequency. But due to the lack of crafting and soldering accuracy and connection error, the experiment was not a success. As a result, the required result, according to fig 1.5, was not achieved.

However, the objectives of understanding astable multivibrator and implementing it on PCB were successfully achieved.

10 Conclusion

The construction of the astable multivibrator circuit using the 555 timer IC, while theoretically straightforward, faced practical challenges during the soldering and assembling. This highlights the importance of good soldering techniques and accurate component placement. Future efforts should be given on good soldering, checking the components, and electrical testing of every connections. Identifying and correcting these errors is crucial for achieving the desired output from the astable multivibrator.

In conclusion, the 555 timer IC, configured as an astable multivibrator, functions as a self-triggering oscillator. It continuously generates a square wave output, highlighting its utility in waveform generation applications.